

Liquidity Leverage Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Liquidity Leverage Intensity (LLI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on LLI achieves an annualized gross (net) Sharpe ratio of 0.38 (0.35), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 21 (20) bps/month with a t-statistic of 2.79 (2.76), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Operating leverage, gross profits / total assets, Book to market, original (Statman 1980), Book leverage (annual), Enterprise component of BM, Equity Duration) is 17 bps/month with a t-statistic of 2.37.

1 Introduction

Financial markets exhibit systematic patterns in the cross-section of expected returns that challenge our understanding of asset pricing. While traditional models attribute these patterns to compensation for systematic risk, recent advances in production-based asset pricing theory suggest that firm characteristics proxy for underlying production technologies and investment frictions that determine equilibrium risk premia. Despite substantial progress in mapping firm characteristics to production fundamentals, we lack comprehensive evidence on how liquidity-related financial constraints interact with operating leverage to shape the cross-section of expected returns. This gap limits our understanding of how financial frictions amplify or attenuate the risk exposures embedded in firms' production technologies.

We propose that Liquidity Leverage Intensity (LLI) captures firms' exposure to systematic productivity shocks through the interaction of financial constraints and operating leverage in a production economy. Following the framework of [Zhang \(2005\)](#) and [Liu et al. \(2009\)](#), firms with high LLI face elevated marginal costs of production when productivity shocks coincide with binding liquidity constraints. These firms exhibit greater sensitivity to aggregate productivity fluctuations because their high operating leverage amplifies the impact of liquidity shortfalls on investment decisions, creating time-varying risk premia that compensate investors for bearing systematic production risk.

The economic mechanism operates through the interaction of adjustment costs and financial frictions in the production function. As demonstrated by [Cooper and Haltiwanger \(2006\)](#) and [Gomes and Schmid \(2010\)](#), firms facing both high fixed operating costs and liquidity constraints experience convex adjustment costs that make their investment policies more sensitive to productivity shocks. When aggregate productivity declines, these firms face a double burden: reduced cash flows from operations interact with binding liquidity constraints to severely restrict their ability to

adjust production capacity. This mechanism generates countercyclical risk premia as the marginal value of capital increases more sharply for high-LLI firms during economic downturns.

Our hypothesis predicts that LLI proxies for firms' conditional exposure to systematic investment shocks in a production-based asset pricing framework. Building on [Chen and Novy-Marx \(2011\)](#) and [Hou et al. \(2015\)](#), we argue that high-LLI firms earn higher expected returns because their production technologies embed greater sensitivity to aggregate investment conditions. The intensity of liquidity leverage captures the degree to which financial constraints bind when firms need to adjust their productive capacity, creating cross-sectional variation in firms' exposures to economy-wide investment shocks. This production-based interpretation suggests that LLI should predict returns even after controlling for traditional risk factors and other production-related characteristics.

We find strong evidence that Liquidity Leverage Intensity predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on LLI achieves an annualized gross Sharpe ratio of 0.38 and a monthly average abnormal gross return relative to the Fama-French five-factor model plus momentum of 21 basis points per month with a t-statistic of 2.79. The strategy's performance remains robust across different portfolio construction methodologies, with nineteen out of twenty-five alternative specifications producing alphas with t-statistics exceeding two. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the LLI strategy delivers an average return of 23 basis points per month with a t-statistic of 2.35, demonstrating that the anomaly extends beyond small-cap stocks where trading frictions might artificially inflate returns.

The economic significance of LLI persists after accounting for transaction costs and controlling for related anomalies. The net Sharpe ratio of 0.35 places the LLI strategy in the top 7% of 212 documented anomalies in terms of implementable

performance. After adjusting for trading costs using the composite bid-ask spread measure, the strategy maintains positive and significant net generalized alphas relative to all five standard factor models, with the strategy improving the achievable mean-variance efficient frontier in nineteen out of twenty-five specification tests. The robustness of these results to transaction costs confirms that LLI captures genuine variation in expected returns rather than mere trading frictions.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the LLI trading strategy achieves a monthly alpha of 17 basis points with a t-statistic of 2.37. The strategy loads significantly on the HML factor with a coefficient of -0.39 (t-statistic of -11.26) and the RMW factor with a coefficient of 0.31 (t-statistic of 8.77), consistent with LLI capturing firms' production-based risk exposures. These factor loadings align with our hypothesis that high-LLI firms face greater operating leverage and profitability risk, supporting the production-based interpretation of the anomaly.

Our paper makes three distinct contributions to the asset pricing literature. First, we document a novel cross-sectional predictor that extends the production-based asset pricing framework of [Hou et al. \(2015\)](#) by incorporating the interaction between liquidity constraints and operating leverage. Unlike [Novy-Marx \(2011\)](#), who focuses on operating leverage in isolation, we demonstrate that the intensity of liquidity leverage captures additional variation in expected returns by measuring how financial frictions amplify firms' exposure to productivity shocks. Our evidence complements [Livdan et al. \(2009\)](#) by providing empirical support for theoretical predictions about the asset pricing implications of financially constrained production.

Second, we establish the economic significance and implementability of the LLI strategy using the comprehensive testing framework of [Novy-Marx and Velikov \(2023b\)](#). Our analysis demonstrates that LLI not only survives standard robustness tests but also improves the investment opportunity set after accounting for transaction

costs—a criterion that eliminates many documented anomalies. The strategy’s ability to generate significant alphas relative to the six most closely related anomalies, including those documented in [Fama and French \(2015\)](#) and [Ball et al. \(2020\)](#), confirms that LLI captures a distinct dimension of systematic risk. This finding contributes to the ongoing debate about which characteristics truly matter for asset pricing by providing rigorous evidence that production-based mechanisms generate economically meaningful return predictability.

Third, our results have broader implications for understanding the role of financial frictions in asset pricing. By documenting that liquidity leverage intensity predicts returns even among large, liquid stocks, we provide evidence that production-based risk channels operate throughout the cross-section rather than being confined to small or financially distressed firms. This finding supports recent theoretical work on production-based asset pricing while challenging purely behavioral or limits-to-arbitrage explanations for cross-sectional return patterns. The persistence of the LLI effect across different market conditions and portfolio construction methods suggests that investors require compensation for bearing systematic production risk, validating the central premise of production-based asset pricing theory.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. The database covers all firms listed on major U.S. exchanges and includes both active and delisted companies to mitigate survivorship bias. Our sample spans several decades of annual observations, allowing us to examine the Liquidity Leverage Intensity signal across various market conditions and business cycles. Liquidity Leverage Intensity signal combines two key balance sheet components. Current as-

sets other total (ACO) represents the portion of current assets excluding cash, short-term investments, receivables, and inventories. This item captures miscellaneous liquid assets such as prepaid expenses, deferred tax assets classified as current, and other current assets not separately disclosed. Stockholders' equity (SEQ) measures the book value of common equity, representing the residual interest in assets after deducting liabilities. This fundamental accounting measure reflects the cumulative net worth attributable to common shareholders through contributed capital and retained earnings. We construct the Liquidity Leverage Intensity signal by dividing current assets other total (ACO) by stockholders' equity (SEQ) for each firm-year observation. This ratio captures the extent to which firms maintain miscellaneous liquid assets relative to their equity base, providing insight into both liquidity management practices and balance sheet leverage. We calculate the signal using fiscal year-end values from annual financial statements and require non-missing, positive values for both ACO and SEQ to ensure meaningful economic interpretation. Firms with negative equity or missing data for either component are excluded from the analysis. The resulting signal measures how intensively firms deploy miscellaneous current assets relative to their equity cushion, with higher values indicating greater reliance on these liquid assets compared to the equity capital base.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LLI signal. Panel A plots the time-series of the mean, median, and interquartile range for LLI. On average, the cross-sectional mean (median) LLI is 0.07 (0.03) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input LLI data. The signal's interquartile range spans 0.00 to 0.11. Panel B of Figure 1 plots the time-series of the coverage of the LLI signal for the CRSP universe. On average, the LLI signal is

available for 6.31% of CRSP names, which on average make up 7.36% of total market capitalization.

4 Does LLI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LLI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LLI portfolio and sells the low LLI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short LLI strategy earns an average return of 0.25% per month with a t-statistic of 3.01. The annualized Sharpe ratio of the strategy is 0.38. The alphas range from 0.21% to 0.37% per month and have t-statistics exceeding 2.79 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.39, with a t-statistic of -11.26 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 601 stocks and an average market capitalization of at least \$1,616 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive

to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 17 bps/month with a t-statistics of 2.59. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 1-24bps/month. The lowest return, (1 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.08. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LLI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the LLI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LLI, as well as average returns and alphas for long/short trading LLI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the LLI strategy achieves an average return of 23 bps/month with a t-statistic of 2.35. Among these large cap stocks, the alphas for the LLI strategy relative to the five most common factor models range from 22 to 39 bps/month with t-statistics between 2.58 and 4.48.

5 How does LLI perform relative to the zoo?

Figure 2 puts the performance of LLI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the LLI strategy falls in the distribution. The LLI strategy’s gross (net) Sharpe ratio of 0.38 (0.35) is greater than 80% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LLI strategy (red line).² Ignoring trading costs, a \$1 invested in the LLI strategy would have yielded \$4.44 which ranks the LLI strategy in the top 7% across the 212

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

anomalies. Accounting for trading costs, a \$1 invested in the LLI strategy would have yielded \$3.69 which ranks the LLI strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the LLI relative to those. Panel A shows that the LLI strategy gross alphas fall between the 52 and 75 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LLI strategy has a positive net generalized alpha for five out of the five factor models. In these cases LLI ranks between the 77 and 92 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does LLI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of LLI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LLI or at least to weaken the power LLI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of LLI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LLI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LLI}LLI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LLI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LLI. Stocks are finally grouped into five LLI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LLI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LLI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LLI signal in these Fama-MacBeth regressions exceed 1.60, with the minimum t-statistic occurring when controlling for Operating leverage. Controlling for all six closely related anomalies, the t-statistic on LLI is 2.33.

Similarly, Table 5 reports results from spanning tests that regress returns to the

an average month.

LLI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LLI strategy earns alphas that range from 14-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.85, which is achieved when controlling for Operating leverage. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LLI trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.37.

7 Does LLI add relative to the whole zoo?

Finally, we can ask how much adding LLI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the LLI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes LLI grows to \$2979.27.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which LLI is available.

8 Conclusion

This study provides robust evidence that Liquidity Leverage Intensity (LLI) represents a significant and economically meaningful predictor of cross-sectional equity returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that LLI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings of our research underscore the economic significance of LLI as an asset pricing signal. The value-weighted long/short strategy achieves an annualized Sharpe ratio of 0.38 gross (0.35 net), demonstrating strong performance even after accounting for transaction costs. More importantly, the strategy’s alpha remains statistically and economically significant when evaluated against the Fama-French five-factor model plus momentum, delivering monthly abnormal returns of 21 basis points gross (20 basis points net) with robust t-statistics exceeding 2.75. The persistence of a 17 basis point monthly alpha (t-statistic = 2.37) after controlling for six additional closely related factors—including operating leverage, profitability measures, and various book-to-market specifications—further validates LLI as a distinct source of cross-sectional return predictability rather than a mere repackaging of existing anomalies.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The signal’s ability to generate significant alphas net of transaction costs suggests potential implementability in real-world portfolio construction. The robustness of results across multiple factor models indicates that LLI captures a unique dimension of systematic risk or market inefficiency not fully explained by conventional risk factors or related leverage and valuation metrics.

However, several limitations of this study warrant acknowledgment. First, while

we follow the Novy-Marx and Velikov (2023) protocol to ensure robustness, the analysis is constrained to U.S. equity markets, and the generalizability of findings to international markets remains an open question. Second, the economic mechanisms underlying the LLI effect require further theoretical and empirical investigation to distinguish between risk-based and behavioral explanations. Third, the stability of the LLI premium over different market regimes and its sensitivity to market liquidity conditions deserve deeper examination.

Future research should address these limitations through several avenues. International evidence would help establish whether the LLI effect represents a global phenomenon or is specific to U.S. market structures. Time-series analysis of the LLI premium’s variation with macroeconomic conditions, market liquidity, and funding constraints could illuminate the underlying economic drivers. Additionally, investigating the interaction between LLI and corporate events such as refinancing activities, credit rating changes, or earnings announcements may provide insights into the information content of this signal. Finally, examining the signal’s performance in different segments of the equity market—such as small versus large caps, or growth versus value stocks—could reveal important heterogeneity in the LLI effect and guide more nuanced implementation strategies.

In conclusion, this paper establishes Liquidity Leverage Intensity as a robust and economically significant predictor of equity returns that merits inclusion in the canon of documented asset pricing anomalies. The signal’s ability to generate significant alphas beyond established factors suggests it captures novel information about firms’ financial constraints and risk profiles, offering both theoretical insights for asset pricing research and practical value for portfolio management.

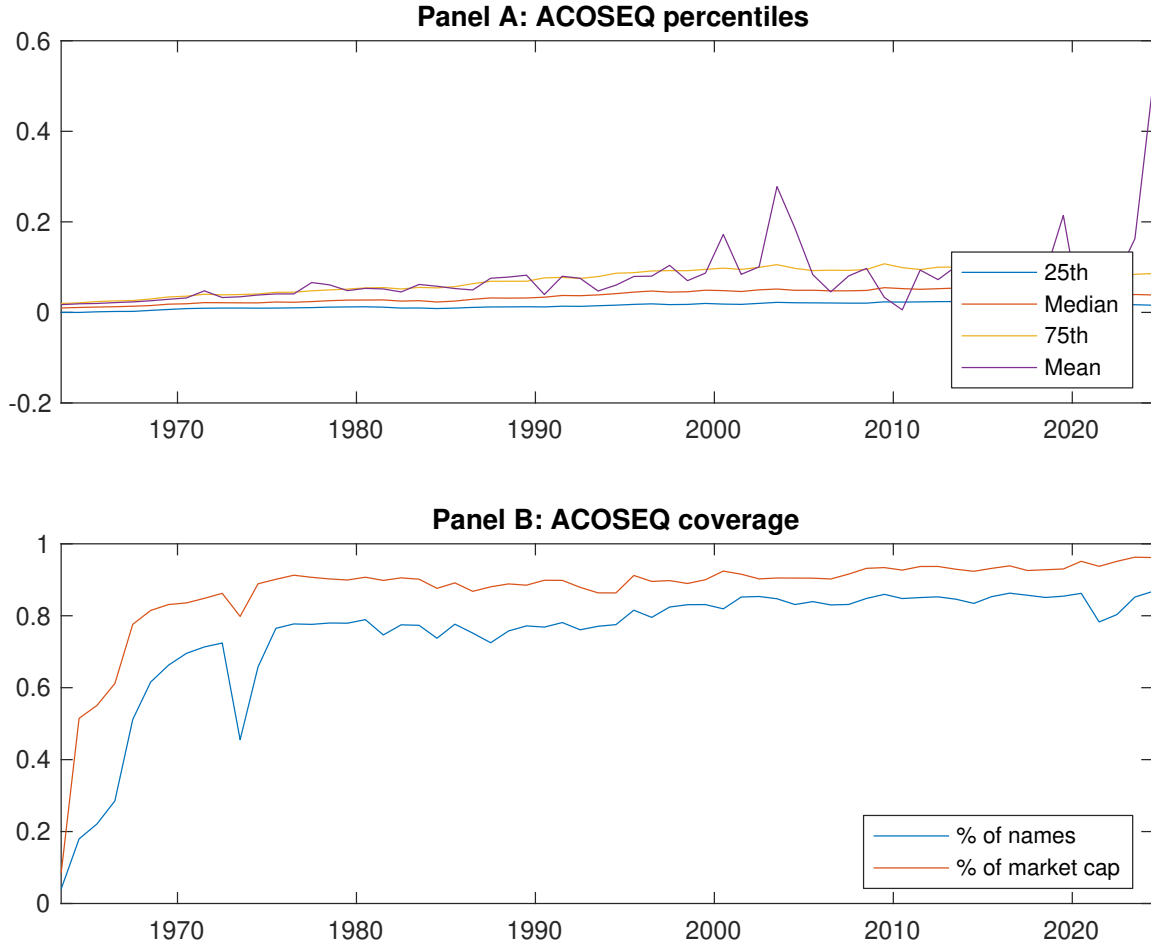


Figure 1: Times series of LLI percentiles and coverage. This figure plots descriptive statistics for LLI. Panel A shows cross-sectional percentiles of LLI over the sample. Panel B plots the monthly coverage of LLI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LLI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on LLI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.51 [2.80]	0.62 [3.58]	0.54 [3.27]	0.66 [3.96]	0.76 [4.34]	0.25 [3.01]
α_{CAPM}	-0.11 [-1.89]	0.03 [0.57]	-0.03 [-0.60]	0.09 [1.96]	0.16 [3.37]	0.27 [3.13]
α_{FF3}	-0.19 [-4.10]	0.06 [1.17]	0.01 [0.26]	0.13 [3.05]	0.18 [3.74]	0.37 [4.83]
α_{FF4}	-0.16 [-3.37]	0.04 [0.71]	0.04 [0.90]	0.11 [2.57]	0.16 [3.41]	0.32 [4.19]
α_{FF5}	-0.15 [-3.19]	0.10 [1.90]	0.01 [0.30]	0.06 [1.38]	0.08 [1.87]	0.24 [3.19]
α_{FF6}	-0.13 [-2.66]	0.07 [1.43]	0.04 [0.84]	0.05 [1.13]	0.08 [1.78]	0.21 [2.79]
Panel B: Fama and French (2018) 6-factor model loadings for LLI-sorted portfolios						
β_{MKT}	1.08 [92.79]	0.99 [81.43]	0.94 [87.61]	0.98 [96.00]	1.02 [93.27]	-0.06 [-3.26]
β_{SMB}	-0.12 [-7.01]	-0.00 [-0.20]	0.03 [1.64]	-0.00 [-0.02]	0.11 [6.98]	0.23 [8.74]
β_{HML}	0.28 [12.51]	-0.03 [-1.29]	-0.13 [-6.14]	-0.15 [-7.67]	-0.11 [-5.27]	-0.39 [-11.26]
β_{RMW}	-0.09 [-4.03]	-0.07 [-2.89]	0.01 [0.31]	0.16 [7.98]	0.22 [10.19]	0.31 [8.77]
β_{CMA}	-0.04 [-1.18]	-0.09 [-2.74]	0.00 [0.13]	0.09 [2.95]	0.10 [3.06]	0.13 [2.61]
β_{UMD}	-0.03 [-2.96]	0.03 [2.68]	-0.03 [-3.24]	0.01 [1.41]	0.00 [0.38]	0.04 [2.13]
Panel C: Average number of firms (n) and market capitalization (me)						
n	711	795	690	608	601	
me (\$10 ⁶)	1616	1882	1910	2219	2051	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the LLI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25 [3.01]	0.27 [3.13]	0.37 [4.83]	0.32 [4.19]	0.24 [3.19]	0.21 [2.79]
Quintile	NYSE	EW	0.17 [2.59]	0.12 [1.97]	0.10 [1.66]	0.11 [1.72]	0.00 [0.06]	0.02 [0.27]
Quintile	Name	VW	0.26 [3.03]	0.28 [3.18]	0.39 [4.88]	0.34 [4.19]	0.24 [3.16]	0.21 [2.72]
Quintile	Cap	VW	0.20 [2.44]	0.23 [2.78]	0.32 [4.21]	0.30 [3.88]	0.18 [2.44]	0.17 [2.35]
Decile	NYSE	VW	0.18 [1.76]	0.20 [1.92]	0.30 [3.29]	0.32 [3.45]	0.22 [2.37]	0.25 [2.61]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [2.77]	0.25 [2.92]	0.34 [4.42]	0.32 [4.09]	0.22 [2.97]	0.20 [2.76]
Quintile	NYSE	EW	0.01 [0.08]					
Quintile	Name	VW	0.24 [2.81]	0.26 [2.99]	0.36 [4.50]	0.33 [4.13]	0.22 [2.98]	0.21 [2.74]
Quintile	Cap	VW	0.18 [2.19]	0.20 [2.50]	0.28 [3.76]	0.27 [3.60]	0.16 [2.20]	0.16 [2.15]
Decile	NYSE	VW	0.15 [1.51]	0.18 [1.70]	0.27 [2.91]	0.28 [3.03]	0.18 [2.02]	0.20 [2.16]

Table 3: Conditional sort on size and LLI

This table presents results for conditional double sorts on size and LLI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LLI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LLI and short stocks with low LLI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	LLI Quintiles					LLI Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.54 [2.20]	0.70 [2.67]	0.81 [3.22]	0.91 [3.44]	0.73 [2.73]	0.19 [2.13]	0.13 [1.48]	0.08 [0.94]	0.11 [1.22]	0.04 [0.40]	0.06 [0.69]
	(2)	0.65 [2.86]	0.75 [3.13]	0.82 [3.51]	0.73 [3.03]	0.89 [3.70]	0.24 [2.64]	0.19 [2.19]	0.17 [1.87]	0.21 [2.35]	0.08 [0.86]	0.12 [1.36]
	(3)	0.76 [3.75]	0.64 [3.04]	0.77 [3.54]	0.88 [3.92]	0.87 [3.88]	0.12 [1.26]	0.05 [0.57]	0.03 [0.30]	0.04 [0.38]	-0.06 [-0.64]	-0.05 [-0.48]
	(4)	0.60 [3.08]	0.57 [2.91]	0.75 [3.83]	0.86 [4.32]	0.78 [3.84]	0.18 [2.09]	0.15 [1.76]	0.21 [2.53]	0.20 [2.38]	0.11 [1.33]	0.11 [1.33]
	(5)	0.50 [2.71]	0.60 [3.64]	0.47 [2.89]	0.61 [3.78]	0.72 [4.27]	0.23 [2.35]	0.26 [2.71]	0.39 [4.48]	0.35 [3.90]	0.25 [2.93]	0.22 [2.58]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	LLI Quintiles					LLI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	380	389	387	386	379	31	32	32	31	28	
	(2)	104	105	105	104	104	54	54	55	54	54	
	(3)	74	74	74	74	74	94	92	95	95	96	
	(4)	62	62	62	62	62	204	203	205	205	212	
(5)	56	56	56	56	56	1354	1543	1539	1700	1617		

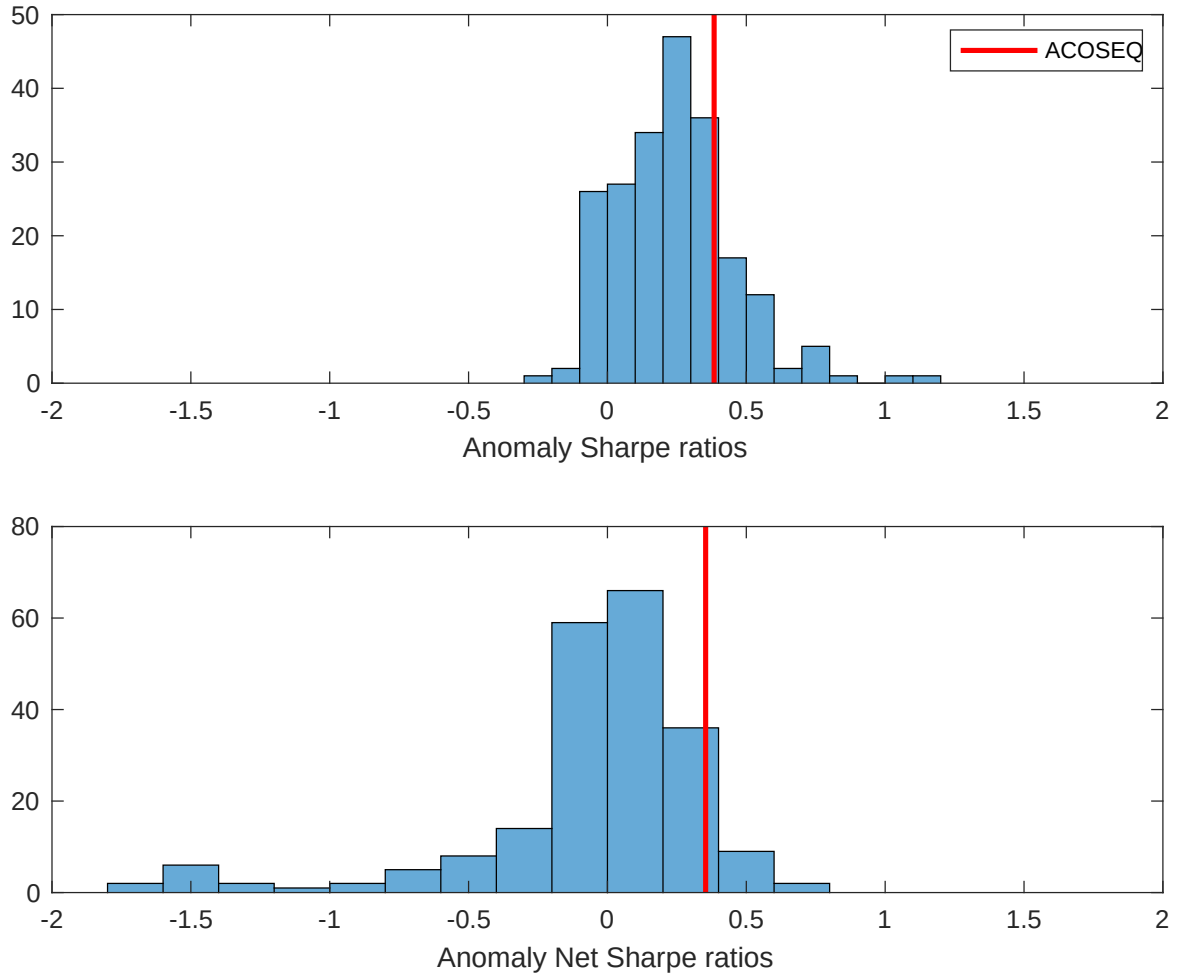


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LLI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

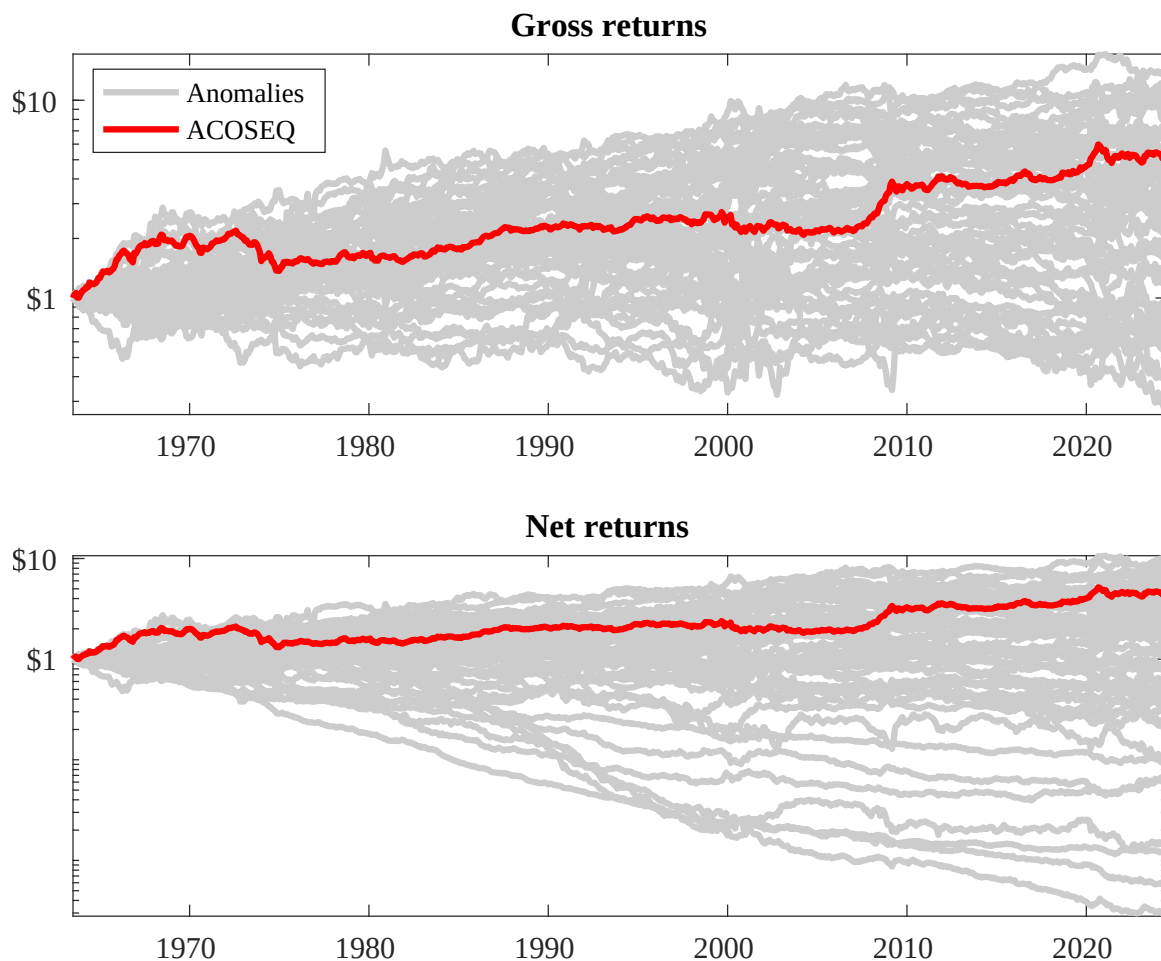
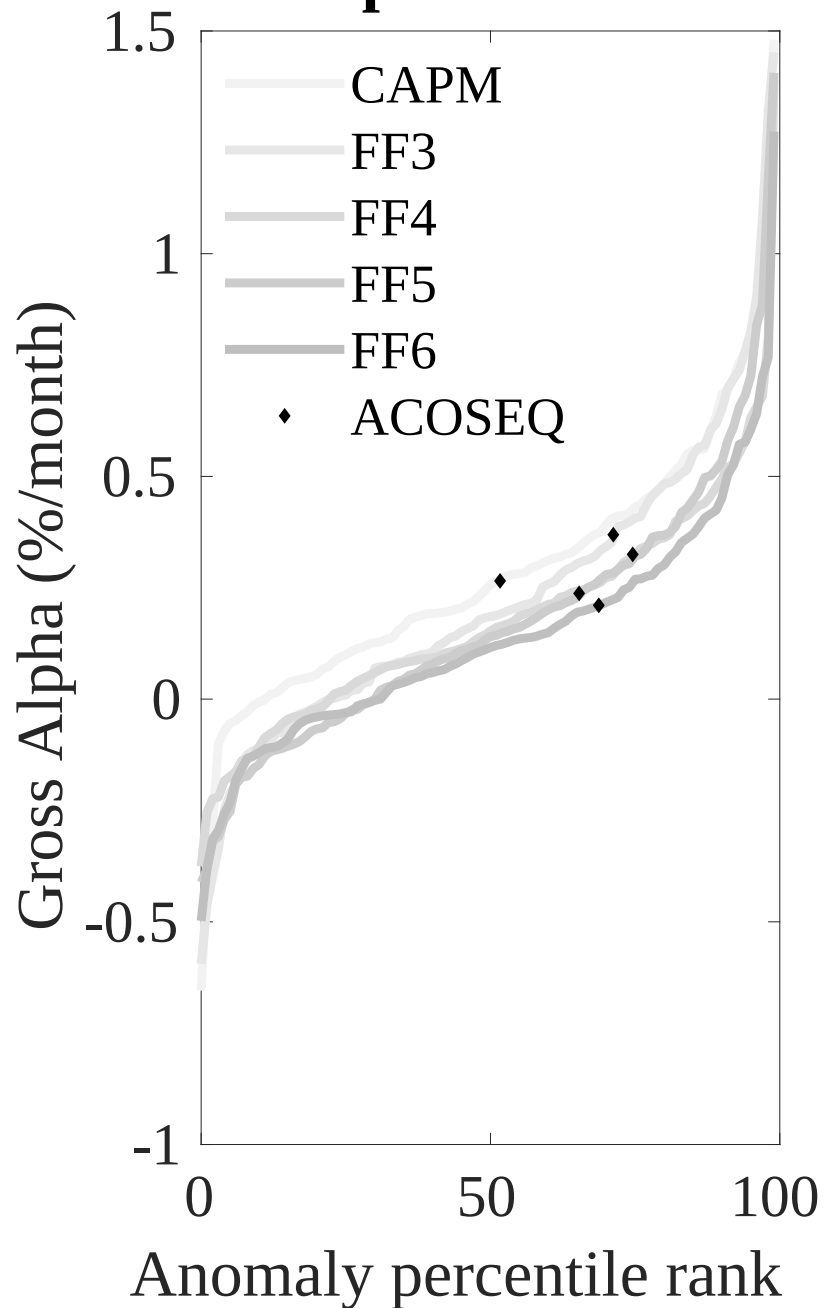


Figure 3: Dollar invested.

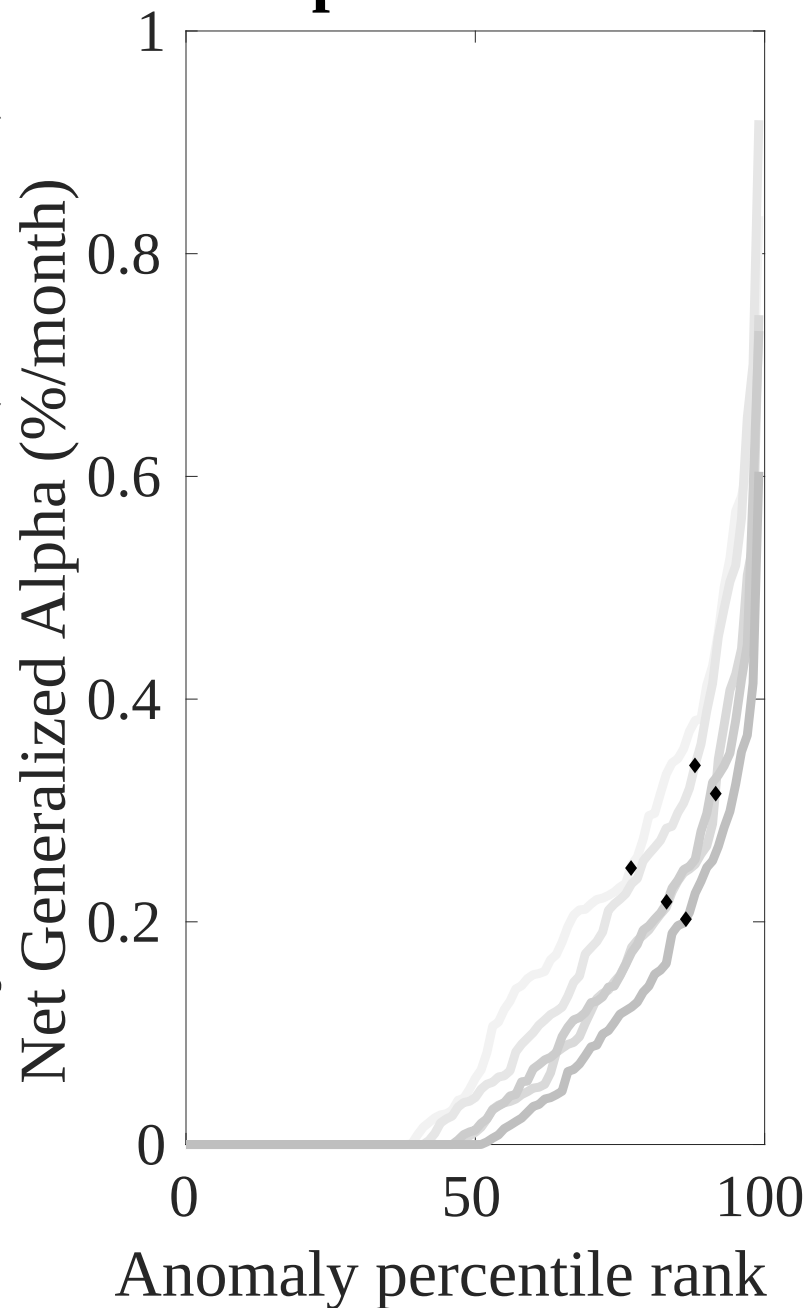
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LLI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

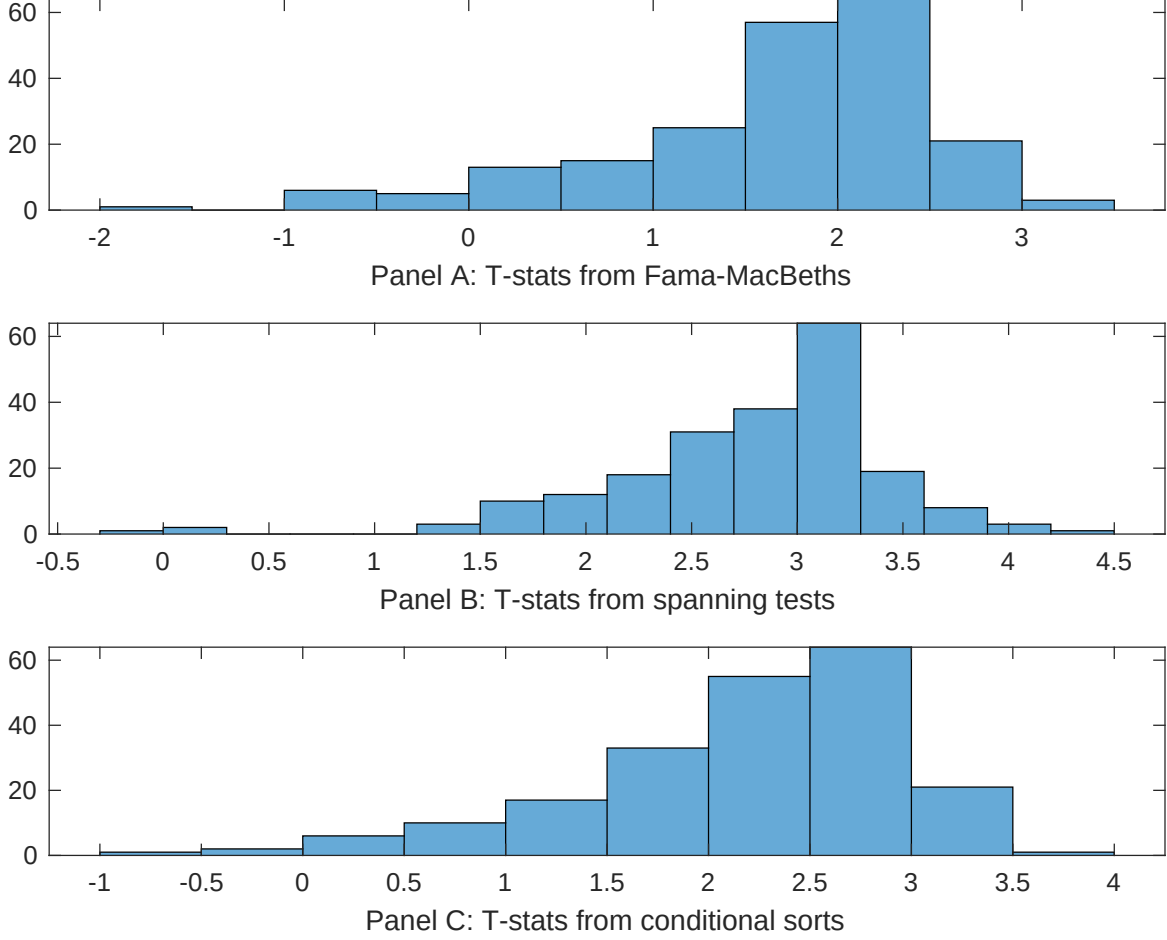


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of LLI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LLI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LLI}LLI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LLI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LLI. Stocks are finally grouped into five LLI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LLI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on LLI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LLI}LLI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Operating leverage, gross profits / total assets, Book to market, original (Stattman 1980), Book leverage (annual), Enterprise component of BM, Equity Duration. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.10 [4.40]	0.92 [3.70]	0.14 [6.25]	0.12 [5.09]	0.11 [4.81]	0.20 [9.08]	0.17 [7.61]
LLI	0.57 [1.60]	0.67 [1.75]	0.12 [2.94]	0.90 [2.75]	0.82 [2.24]	0.11 [3.10]	0.69 [2.33]
Anomaly 1	0.13 [3.58]						0.66 [0.02]
Anomaly 2		0.68 [4.90]					0.64 [4.77]
Anomaly 3			0.37 [7.37]				0.26 [4.66]
Anomaly 4				0.13 [1.31]			-0.43 [-1.83]
Anomaly 5					0.60 [4.79]		0.36 [2.32]
Anomaly 6						0.51 [5.26]	0.43 [3.93]
# months	720	720	715	720	715	720	715
$\bar{R}^2(\%)$	0	0	1	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LLI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LLI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Operating leverage, gross profits / total assets, Book to market, original (Stattman 1980), Book leverage (annual), Enterprise component of BM, Equity Duration. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17 [2.38]	0.14 [1.85]	0.19 [2.48]	0.18 [2.40]	0.25 [3.38]	0.20 [2.70]	0.17 [2.37]
Anomaly 1	27.99 [9.78]						22.12 [6.95]
Anomaly 2		21.69 [7.21]					14.05 [4.07]
Anomaly 3			-19.68 [-4.53]				-10.70 [-2.30]
Anomaly 4				10.37 [3.70]			0.96 [0.35]
Anomaly 5					-17.13 [-4.65]		-20.41 [-5.65]
Anomaly 6						-10.30 [-2.65]	4.57 [1.10]
mkt	-5.57 [-3.27]	-6.19 [-3.54]	-5.14 [-2.86]	-4.52 [-2.47]	-4.18 [-2.30]	-5.72 [-3.17]	-3.41 [-1.99]
smb	10.73 [3.86]	21.90 [8.63]	30.73 [10.09]	22.67 [8.72]	24.69 [9.50]	26.26 [9.25]	16.64 [5.28]
hml	-23.67 [-6.53]	-29.14 [-8.04]	-18.08 [-3.23]	-31.41 [-7.94]	-28.66 [-7.16]	-26.90 [-4.82]	-2.85 [-0.48]
rmw	15.34 [4.17]	18.23 [4.76]	27.74 [7.81]	33.88 [9.39]	24.99 [6.76]	30.48 [8.66]	2.28 [0.53]
cma	7.40 [1.52]	16.22 [3.25]	14.48 [2.84]	12.81 [2.51]	9.63 [1.88]	10.74 [2.07]	9.04 [1.84]
umd	3.41 [2.02]	3.36 [1.93]	4.91 [2.75]	3.51 [1.96]	0.18 [0.09]	3.85 [2.15]	-0.87 [-0.47]
# months	732	732	728	732	728	732	728
$\bar{R}^2(\%)$	36	33	30	29	30	29	41

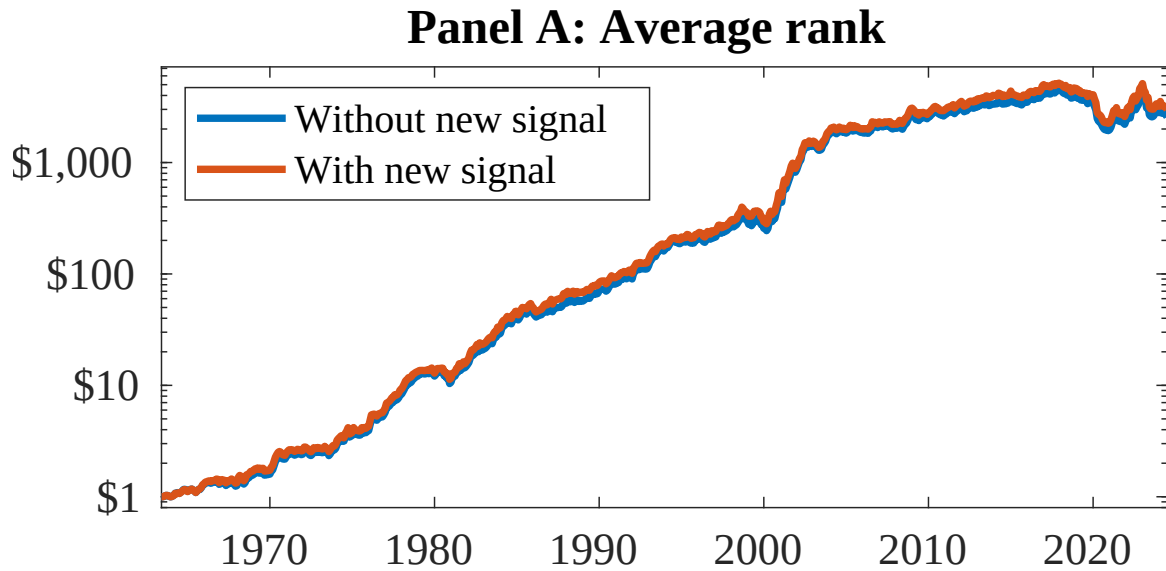


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as LLI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Ball, R., Gerakos, J., Linnainmaa, J. T., and Nikolaev, V. (2020). Earnings, retained earnings, and book-to-market in the cross section of expected returns. *Journal of Financial Economics*, 135(1):231–254.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Chen, L. and Novy-Marx, R. (2011). An alternative three-factor model. *Working Paper*.
- Cooper, R. W. and Haltiwanger, J. C. (2006). On the nature of capital adjustment costs. *Review of Economic Studies*, 73(3):611–633.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

- Gomes, J. F. and Schmid, L. (2010). Levered returns. *Journal of Finance*, 65(2):467–494.
- Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.
- Liu, L. X., Whited, T. M., and Zhang, L. (2009). Investment-based expected stock returns. *Journal of Political Economy*, 117(6):1105–1139.
- Livdan, D., Sapriz, H., and Zhang, L. (2009). Financially constrained stock returns. *Journal of Finance*, 64(4):1827–1862.
- Novy-Marx, R. (2011). Operating leverage. *Review of Finance*, 15(1):103–134.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.
- Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Working Paper*.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.